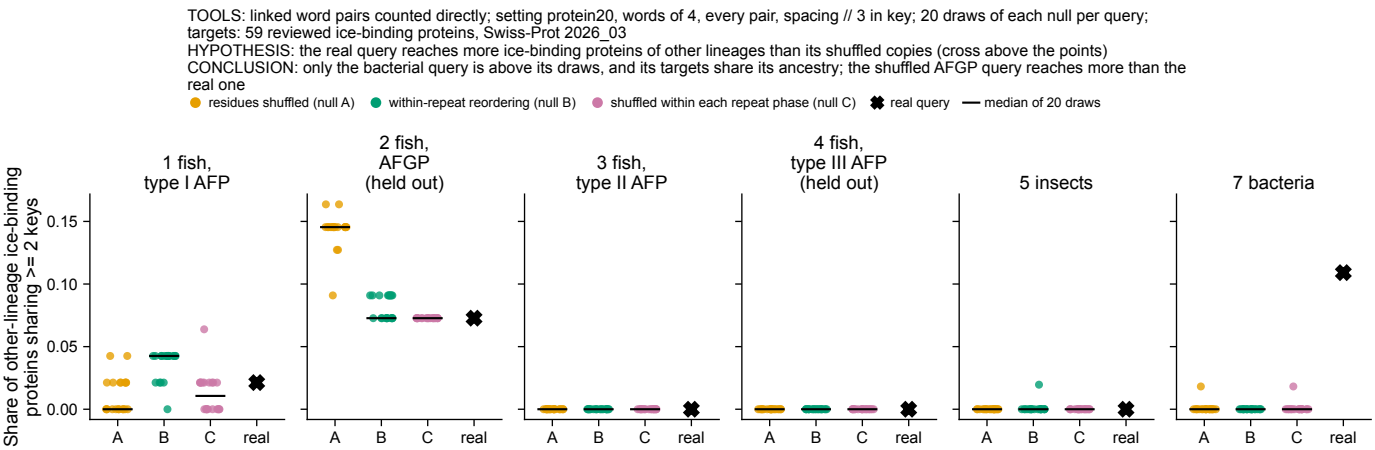
TOOLS: linked word pairs counted directly; setting protein20, words of 4, every pair, spacing // 3 in key; 20 draws of each null per query; targets: 59 reviewed ice-binding proteins, Swiss-Prot 2026_03
HYPOTHESIS: the real query reaches more ice-binding proteins of other lineages than its shuffled copies (cross above the points)
CONCLUSION: only the bacterial query is above its draws, and its targets share its ancestry; the shuffled AFGP query reaches more than the real one
residues shuffled (null A)
within-repeat reordering (null B)
shuffled within each repeat phase (null C)
real query
median of 20 draws
Share of other-lineage ice-binding proteins sharing >= 2 keys
0.15
0.10
0.05
0.00
1 fish, type I AFP
2 fish, AFGP (held out)
3 fish, type II AFP
4 fish, type III AFP (held out)
5 insects
7 bacteria
A
B
C
real